EDRA: An Epistemically Disciplined Reasoning Architecture for Grounded Symbolic Inference

Jeevan Naidu
2023.jeevann@isu.ac.in

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Abstract

Context: Contemporary reasoning systems face a fundamental tension between inferential coverage and epistemic correctness, often generating plausible but unjustified conclusions when evidence is insufficient.

Problem: Existing symbolic reasoning architectures lack formal mechanisms to treat refusal as a principled epistemic action, leading to unconstrained inference beyond evidential support.

Objective: This research presents EDRA (Epistemically Disciplined Reasoning Architecture), a modular symbolic reasoning system that enforces epistemic discipline through grounding verification, negation-dominant inheritance, and proof obligations.

Method: EDRA implements a six-component architecture (Manas, Buddhi, Chitta, HRE, Ahankara, Sakshin) with formal semantics for beliefs, taxonomic relations, and refusal conditions. We evaluate the system using a 44-query benchmark designed to falsify epistemic guarantees across five categories: direct facts, taxonomic inheritance, negation dominance, grounding discipline, and hypothetical reasoning.

Results: EDRA achieves 84% precision with 85% restraint rate, demonstrating zero false refusals (100% restraint accuracy) on grounding discipline queries. The system correctly refuses all 8 ungrounded queries while maintaining 91% overall accuracy on answerable queries. Negation dominance and inheritance accuracy both reach 82%, with failures occurring primarily in multiple inheritance conflicts.

Contribution: This work establishes refusal as a first-class epistemic outcome with formal semantics, introduces relation frame theory for compositional inference control, and demonstrates that epistemic discipline can substantially reduce false positives without sacrificing transparency.

Significance: EDRA provides a reasoning substrate for applications where epistemic correctness outweighs coverage, with applications in safety-critical systems, knowledge verification, and hybrid neural-symbolic architectures.
Keywords: Epistemic discipline, symbolic reasoning, refusal semantics, non-monotonic logic, knowledge representation, grounding verification

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1 Introduction

1.1 Background and Context

Reasoning systems operate at the intersection of knowledge representation and inference, transforming stored facts into derived conclusions. This fundamental capability underlies applications from question answering to automated decision-making. However, contemporary systems face a persistent challenge: the tension between *inferential coverage*—the breadth of queries a system can answer—and *epistemic correctness*—the accuracy and justifiability of those answers.

Systems optimized for coverage attempt to answer as many queries as possible, often extending inference beyond what is justified by their knowledge base. Statistical language models generate fluent responses even when evidence is weak or absent [18]. Classical symbolic systems permit unconstrained composition, silently overgeneralizing from stored facts. In both paradigms, systems appear knowledgeable while exceeding their epistemic authority.

The cost of this overreach is measurable: false positives in medical diagnosis systems, hallucinated citations in information retrieval, and confabulated reasoning in automated assistants. These failures share a common root: the absence of formal mechanisms to recognize and communicate epistemic boundaries.

1.2 Problem Statement

Current symbolic reasoning architectures lack principled mechanisms to:

1. **Detect inference beyond evidential support:** Systems perform derivations without verifying that predicates are grounded in stored or derivable knowledge.

2. **Handle conflicting evidence formally:** When inherited positives conflict with direct negations, most systems either select arbitrarily or fail silently.

3. **Treat refusal as a valid epistemic action:** Uncertainty is encoded implicitly (e.g., low confidence scores) or suppressed entirely, rather than being an explicit, explainable outcome.

4. **Provide proof obligations for asserted conclusions:** Derived answers often lack traceable derivations, making verification and debugging intractable.

Research Question: Can a symbolic reasoning system enforce epistemic discipline—restricting inference to justified knowledge while explicitly acknowledging uncertainty—without sacrificing modularity, transparency, or formal traceability?

1.3 Significance of the Study

This research addresses a foundational challenge in knowledge representation: *how to build reasoning systems that know their limits.* The significance is threefold:

Theoretical Contribution: We formalize refusal as a first-class epistemic outcome with explicit semantics, metrics, and proof obligations. This extends classical knowledge representation by treating "I don't know" as rigorously as "yes" or "no."

Practical Impact: Epistemically disciplined systems are critical for safety-critical applications (medical diagnosis, legal reasoning, autonomous systems) where false positives carry high cost and transparency is non-negotiable.

Architectural Innovation: EDRA's modular design separates perception, memory, reasoning, orchestration, hypothetical evaluation, and meta-observation, enabling independent verification and replacement of components.

1.4 Thesis Statement

We argue that epistemic discipline—the property of restricting inference to justified knowledge while explicitly acknowledging uncertainty—can be achieved in symbolic reasoning systems through three formal mechanisms: grounding verification, negation-dominant inheritance, and proof obligations.

We demonstrate that these mechanisms, when implemented in a modular architecture, enable a system to maintain high precision on answerable queries while refusing confidently when evidence is insufficient. The trade-off is restricted expressiveness: EDRA sacrifices coverage and learning capability to guarantee epistemic correctness.

1.5 Overview of Paper Organization

This paper is structured as follows:

Section 2 (Literature Review): Surveys knowledge representation formalisms, non-monotonic reasoning, and refusal mechanisms in AI systems, identifying gaps that EDRA addresses.

Section 3 (Methodology): Describes EDRA's architecture, formal model, and implementation. Introduces belief representation, taxonomic semantics, grounding conditions, negation dominance, relation frames, and refusal semantics.

Section 4 (System Components): Details the six architectural components (Manas, Buddhi, Chitta, HRE, Ahankara, Sakshin) and their formal interfaces.

Section 5 (Evaluation Methodology): Presents the benchmark design, query categorization, expected behavior functions, and correctness criteria.

Section 6 (Results): Reports quantitative results, category-specific performance, error analysis, and restraint discipline metrics.

Section 7 (Discussion): Interprets findings, relates EDRA to monotonic/non-monotonic logic, analyzes computational complexity, discusses limitations, and positions EDRA within the knowledge representation landscape.

Section 8 (Conclusion): Summarizes contributions, restates thesis, and outlines future research directions.

2 Literature Review

2.1 Knowledge Representation Formalisms

Knowledge representation (KR) systems formalize the storage and manipulation of structured knowledge. Classical approaches include:

First-Order Logic (FOL): Provides expressive power through quantifiers, predicates, and functions. However, unrestricted FOL inference is undecidable, and standard implementations lack explicit refusal mechanisms [1].

Description Logics (DL): Restrict FOL to decidable fragments using concepts, roles, and terminological axioms. DL systems like OWL support open-world reasoning but struggle with negation-as-failure and explicit uncertainty [2].

Semantic Networks: Represent knowledge as graphs with typed edges (IS-A, HAS-A, etc.). Early systems like KL-ONE pioneered inheritance reasoning but lacked formal semantics for conflict resolution [3].

Frame-Based Systems: Organize knowledge into frames with slots and fillers, supporting default reasoning. However, frame composition and

negation handling remain ad-hoc [4].

Gap Identified: Existing KR formalisms either sacrifice decidability for expressiveness (FOL) or lack principled refusal semantics (DL, semantic networks, frames). EDRA addresses this by restricting expressiveness while formalizing refusal as a first-class outcome.

2.2 Non-Monotonic Reasoning

Non-monotonic logics permit retraction of conclusions when new information is added, contrasting with classical monotonic logic where:

$$\mathcal{B} \subseteq \mathcal{B}' \implies \mathrm{Th}(\mathcal{B}) \subseteq \mathrm{Th}(\mathcal{B}')$$

Key approaches include:

Default Logic [5]: Introduces defaults of the form "if α and β is consistent, infer γ." Enables reasoning with incomplete information but requires consistency checking.

Circumscription [6]: Minimizes extensions of predicates to capture closed-world assumptions. Computationally expensive and difficult to operationalize.

Autoepistemic Logic [7]: Formalizes reasoning about one's own knowledge. Supports introspective negation but lacks tractable implementations.

Inheritance Networks with Exceptions [8]: Allow specific properties to override inherited defaults. EDRA's negation dominance extends this by formalizing the override mechanism with explicit blocking rules.

Gap Identified: Non-monotonic logics formalize defeasible reasoning but rarely treat refusal as an explicit outcome. EDRA implements non-monotonicity through negation dominance while requiring refusal when evidence is insufficient.

2.3 Refusal and Uncertainty in AI Systems

Few AI systems treat refusal as a principled epistemic action:

Selective Prediction [9]: Machine learning systems abstain from predictions when confidence is low. However, confidence thresholds are heuristic rather than epistemic.

Conformal Prediction [10]: Provides statistically valid confidence sets. Applies to statistical models but not symbolic inference.

Epistemic Planning [11]: Models knowledge and belief in multi-agent systems. Focuses on action planning rather than query answering.

Question Answering with Unanswerable Detection [12]: Neural QA systems detect unanswerable questions. Detection is learned, not formally guaranteed.

Gap Identified: Existing refusal mechanisms are either statistical (lacking formal guarantees) or limited to specific domains. EDRA formalizes refusal with explicit conditions, metrics, and proof obligations applicable to symbolic inference.

2.4 Grounding in Symbolic Systems

Grounding ensures symbols refer to entities with defined semantics:

Symbol Grounding Problem [13]: How do symbols acquire meaning? Classical symbolic AI often assumes grounding is external.

Perceptual Grounding [14]: Grounds symbols in sensory data. EDRA extends this to *epistemic grounding*—ensuring predicates are justified by stored or derivable knowledge.

Gap Identified: Most symbolic systems lack runtime grounding verification. EDRA enforces grounding checks at query time, refusing when predicates are ungrounded.

2.5 Modular Cognitive Architectures

Cognitive architectures decompose reasoning into functional components:

SOAR [15]: Implements production rules with chunking for learning. Monolithic working memory limits modularity.

ACT-R [16]: Models declarative and procedural knowledge separately. Focuses on cognitive fidelity rather than epistemic discipline.

Sigma [17]: Unifies perception, memory, and reasoning in graphical models. Probabilistic but lacks symbolic grounding guarantees.

Gap Identified: Existing architectures prioritize cognitive plausibility or learning. EDRA prioritizes epistemic correctness through modular separation of concerns with explicit interfaces.

2.6 Identifying the Research Gap

Despite advances in knowledge representation, non-monotonic reasoning, and cognitive architectures, **no existing system combines**:

1. Formal grounding verification at query time
2. Negation-dominant inheritance with explicit blocking semantics

3. Refusal as a first-class outcome with proof obligations
4. Modular architecture enabling independent component verification
5. Explicit metrics for epistemic discipline (restraint rate, precision, false positive rate)

EDRA fills this gap by integrating these mechanisms into a unified reasoning substrate with formal semantics and empirical validation.

3 Methodology

3.1 Research Design

This research follows a **constructive methodology** in symbolic AI: we design, implement, and evaluate a formal reasoning system with explicit epistemic constraints. The approach is:

Design Phase: Formalize belief representation, grounding conditions, inheritance semantics, refusal conditions, and proof obligations using set theory and formal logic.

Implementation Phase: Develop a modular Python implementation with six independent components and explicit interfaces.

Evaluation Phase: Construct a benchmark designed to falsify epistemic guarantees, measure performance across query categories, and analyze failure modes.

3.2 Epistemic Discipline as a Design Objective

We define **epistemic discipline** as the property of a reasoning system that restricts inference to justified knowledge while explicitly acknowledging uncertainty. An epistemically disciplined system satisfies:

1. **Grounding Constraint:** Inference is permitted only over predicates grounded in stored or derivable knowledge.
2. **Negation Dominance:** Negative evidence dominates positive inheritance, implementing defeasible reasoning.
3. **Proof Obligation:** Every non-refused conclusion admits an explicit derivation traceable to stored beliefs.
4. **Explicit Refusal:** When evidence is insufficient, the system refuses with a formal justification.

In EDRA, refusal is not an error or fallback—it is a formal epistemic assertion that the system lacks sufficient justification to answer a query.

3.3 Data Collection Methods

Belief Base Construction: The system is populated with 32 beliefs covering:

- Taxonomic relations: 8 IS-A relations forming a 3-level taxonomy (e.g., penguin $\prec$ bird $\prec$ animal)
- Property assertions: 16 beliefs about functional properties (produces_milk, can_fly, is_aquatic)
- Explicit negations: 8 negations overriding inherited positives (e.g., penguin CANNOT fly)

Benchmark Query Collection: 44 queries systematically probe epistemic boundaries:

- 10 direct fact queries (exact matches in belief base)
- 11 inheritance queries (require taxonomic traversal)
- 11 negation dominance queries (test blocking of inherited positives)
- 8 grounding discipline queries (ungrounded predicates, expected refusal)
- 4 hypothetical queries (counterfactual reasoning with isolation)

Expected Behavior Specification: Each query is manually labeled with expected behavior (POSITIVE, NEGATIVE, REFUSED) based on formal semantics.

3.4 Participants and Sample

N/A: This is a formal systems evaluation, not a human subjects study. The "participants" are queries, and the "sample" is the benchmark set.

3.5 Data Analysis Techniques

Categorical Accuracy: For each query category $c \in \{\text{direct}, \text{inherit}, \text{negation}, \text{ground}, \text{hypo}\}$:

$$A_c = \frac{\sum_{q \in \mathcal{Q}_c} \nVdash[\text{Answer}(q) = \text{Expected}(q)]}{|\mathcal{Q}_c|}$$

Restraint Metrics:

$$\text{Restraint Rate } \rho = \frac{|\{q \mid \text{Answer}(q) = \text{REFUSED}\}|}{|Q|} \tag{1}$$

$$\text{Precision } \pi = \frac{|\{q \in Q_{\text{answered}} \mid \text{Answer}(q) = \text{Expected}(q)\}|}{|Q_{\text{answered}}|} \tag{2}$$

$$\text{FPR} = \frac{|\{q \in Q_{\text{answered}} \mid \text{Answer}(q) \neq \text{Expected}(q)\}|}{|Q_{\text{answered}}|} \tag{3}$$

Failure Analysis: Manual inspection of incorrect answers to identify failure modes (multiple inheritance conflicts, deep taxonomy chains, frame ambiguity).

Proof Tracing: For each non-refused answer, verify that a valid proof Π exists using inference rules (DirectMatch, Inheritance, Negation, Transitivity).

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Restraint Metrics:

$$\text{Restraint Rate } \rho = \frac{|\{q \mid \text{Answer}(q) = \text{REFUSED}\}|}{|Q|} \tag{4}$$

$$\text{Precision } \pi = \frac{|\{q \in Q_{\text{answered}} \mid \text{Answer}(q) = \text{Expected}(q)\}|}{|Q_{\text{answered}}|} \tag{5}$$

$$\text{FPR} = \frac{|\{q \in Q_{\text{answered}} \mid \text{Answer}(q) \neq \text{Expected}(q)\}|}{|Q_{\text{answered}}|} \tag{6}$$

Failure Analysis: Manual inspection of incorrect answers to identify failure modes (multiple inheritance conflicts, deep taxonomy chains, frame ambiguity).

Proof Tracing: For each non-refused answer, verify that a valid proof Π exists using inference rules (DirectMatch, Inheritance, Negation, Transitivity).

3.7 Formal Model

3.7.1 Beliefs and Queries

A belief is represented as a tuple:

$$b = (E, P, \nu, \sigma, c, \pi)$$

where:

- $E \subseteq \mathcal{E}$ is a set of entities from entity universe $\mathcal{E}$
- $P \subseteq \mathcal{P}$ is a set of predicates from predicate universe $\mathcal{P}$
- $\nu \in \{\text{POSITIVE}, \text{NEGATIVE}, \text{INTERROGATIVE}\}$ is the polarity
- $\sigma \in \{\text{BELIEVED}, \text{QUESTIONED}, \text{HYPOTHETICAL}, \text{OBSERVED}\}$ is epistemic status
- $c \in [0, 1]$ is a confidence score
- π is provenance metadata (source, timestamp, derivation)

A query is represented as:

$$q = (E_q, P_q, \nu_q)$$

where E_q are the queried entities, P_q are the queried predicates, and $\nu_q =$ INTERROGATIVE.

Let $\mathcal{B}$ denote the belief store and $\prec$ denote the taxonomic (IS-A) relation.

3.7.2 Taxonomic Relation

The taxonomic relation $\prec$ is a partial order on $\mathcal{E}$:

$$e_1 \prec e_2 \implies e_1 \text{ is a subclass/instance of } e_2 \tag{7}$$

$$e_1 \prec e_2 \wedge e_2 \prec e_3 \implies e_1 \prec e_3 \quad \text{(transitivity)} \tag{8}$$

We define the ancestor set:

$$\text{Ancestors}(e) = \{e' \in \mathcal{E} \mid e \prec e'\}$$

And the transitive closure:

$$\prec^* = \prec \cup \{(e_1, e_3) \mid \exists e_2 : e_1 \prec e_2 \wedge e_2 \prec e_3\}$$

3.7.3 Grounding Condition

A predicate $p \in P_q$ is **grounded** for entity $e \in E_q$ if at least one of the following holds:

Direct Grounding:

$$\exists b = (E_b, P_b, \nu_b, \sigma_b, c, \pi) \in \mathcal{B} : e \in E_b \wedge p \in P_b$$

Taxonomic Grounding:

$$\exists e_a \in \text{Ancestors}(e), \exists b \in \mathcal{B} : e_a \in E_b \wedge p \in P_b$$

Simple Property Grounding:

$$|E_q| \leq 2 \wedge \forall e' \in E_q : e' \in \mathcal{E}$$

If none of these conditions hold, the grounding function returns:

$$\text{Ground}(e, p, \mathcal{B}) = \perp \quad \text{(ungrounded)}$$

and inference must refuse.

3.7.4 Negation-Dominant Inheritance

For a query $q = (E_q, P_q, \text{INTERROGATIVE})$, let $e \in E_q$ and $p \in P_q$.

Define the set of inherited negations:

$$\mathcal{N}(e, p) = \{b \in \mathcal{B} \mid \exists e_a \in \text{Ancestors}(e) : e_a \in E_b \wedge p \in P_b \wedge \nu_b = \text{NEGATIVE}\}$$

Negation Dominance Rule:

$$\text{If } \mathcal{N}(e, p) \neq \emptyset, \text{ then Answer}(q) = (\text{NEGATIVE}, \text{inherited from } e_a)$$

This blocks positive inheritance from any path:

$$\mathcal{N}(e, p) \neq \emptyset \implies \neg\exists \text{ positive inheritance of } (e, p)$$

3.7.5 Relation Frame Semantics

Each predicate $p \in \mathcal{P}$ is assigned to a relation frame $F(p) \in \{\text{TAXONOMIC}, \text{FUNCTIONAL}, \text{SPATIAL},$

Frame properties are encoded as:

$$F(p) = (T, I, B)$$

where:

- $T \in \{\text{true}, \text{false}\}$ indicates transitivity
- $I \in \{\text{true}, \text{false}\}$ indicates inheritance
- $B \in \{\text{true}, \text{false}\}$ indicates negation blocking

TAXONOMIC frame: $F(p) = (\text{true}, \text{true}, \text{true})$

$$e_1 \xrightarrow{p} e_2 \wedge e_2 \xrightarrow{p} e_3 \implies e_1 \xrightarrow{p} e_3 \tag{9}$$

$$e_1 \prec e_2 \wedge e_2 \xrightarrow{p} \text{val} \implies e_1 \xrightarrow{p} \text{val} \tag{10}$$

FUNCTIONAL frame: $F(p) = (\text{false}, \text{true}, \text{true})$

$$e_1 \prec e_2 \wedge (e_2, p, \text{POSITIVE}) \in \mathcal{B} \implies (e_1, p, \text{POSITIVE}) \tag{11}$$

$$e_1 \prec e_2 \wedge (e_2, p, \text{NEGATIVE}) \in \mathcal{B} \implies (e_1, p, \text{NEGATIVE}) \tag{12}$$

SPATIAL frame: $F(p) = (\text{true}, \text{false}, \text{false})$

$$e_1 \xrightarrow{p} e_2 \wedge e_2 \xrightarrow{p} e_3 \implies e_1 \xrightarrow{p} e_3 \tag{13}$$

$$\text{No inheritance: } e_1 \prec e_2 \wedge (e_2, p, v) \not\implies (e_1, p, v) \tag{14}$$

STATE frame: $F(p) = (\text{false}, \text{false}, \text{false})$

$$\text{No transitivity, no inheritance} \tag{15}$$

3.7.6 Confidence Propagation

For inherited beliefs, confidence decays with taxonomic distance:

$$c_{\text{inherited}} = c_{\text{source}} \times \delta^d$$

where d is the distance in the taxonomy graph and $\delta \in (0, 1)$ is the decay factor (typically $\delta = 0.9$).

A belief is represented as a tuple:

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where:

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where E_q are the queried entities, P_q are the queried predicates, and $\nu_q =$ INTERROGATIVE.

Let $\mathcal{B}$ denote the belief store and $\prec$ denote the taxonomic (IS-A) relation.

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The taxonomic relation $\prec$ is a partial order on $\mathcal{E}$:

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If none of these conditions hold, the grounding function returns:

$$\text{Ground}(e, p, \mathcal{B}) = \perp \quad \text{(ungrounded)}$$

and inference must refuse.

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Negation Dominance Rule:

$$\text{If } \mathcal{N}(e,p) \neq \emptyset, \text{ then Answer}(q) = (\text{NEGATIVE}, \text{inherited from } e_a)$$

This blocks positive inheritance from any path:

$$\mathcal{N}(e,p) \neq \emptyset \implies \neg\exists \text{ positive inheritance of } (e,p)$$

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Each predicate $p \in \mathcal{P}$ is assigned to a relation frame $F(p) \in \{\text{TAXONOMIC}, \text{FUNCTIONAL}, \text{SPATIAL},$

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- $B \in \{\text{true}, \text{false}\}$ indicates negation blocking

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$$e_1 \xrightarrow{p} e_2 \wedge e_2 \xrightarrow{p} e_3 \implies e_1 \xrightarrow{p} e_3 \tag{18}$$

$$e_1 \prec e_2 \wedge e_2 \xrightarrow{p} \text{val} \implies e_1 \xrightarrow{p} \text{val} \tag{19}$$

FUNCTIONAL frame: $F(p) = (\text{false}, \text{true}, \text{true})$

$$e_1 \prec e_2 \wedge (e_2, p, \text{POSITIVE}) \in \mathcal{B} \implies (e_1, p, \text{POSITIVE}) \tag{20}$$

$$e_1 \prec e_2 \wedge (e_2, p, \text{NEGATIVE}) \in \mathcal{B} \implies (e_1, p, \text{NEGATIVE}) \tag{21}$$

SPATIAL frame: $F(p) = (\text{true}, \text{false}, \text{false})$

$$e_1 \xrightarrow{p} e_2 \wedge e_2 \xrightarrow{p} e_3 \implies e_1 \xrightarrow{p} e_3 \tag{22}$$

$$\text{No inheritance: } e_1 \prec e_2 \wedge (e_2, p, v) \not\Longrightarrow (e_1, p, v) \tag{23}$$

STATE frame: $F(p) = (\text{false}, \text{false}, \text{false})$

$$\text{No transitivity, no inheritance} \tag{24}$$

3.12 Confidence Propagation

For inherited beliefs, confidence decays with taxonomic distance:

$$c_{\text{inherited}} = c_{\text{source}} \times \delta^d$$

where d is the distance in the taxonomy graph and $\delta \in (0, 1)$ is the decay factor (typically $\delta = 0.9$).

4 System Components

EDRA consists of six modular components with explicit interfaces. No component calls another directly; all coordination occurs through the orchestrator (Ahankara).

4.1 Manas: Perception and Parsing

Function: Converts natural language input into structured belief proposals.

Input: String $s \in \Sigma^*$ (natural language)

Output: Belief tuple $b = (E, P, \nu, \sigma, c, \pi)$ or query $q = (E_q, P_q, \nu_q)$

Method: Pattern matching on verb phrases, entity extraction, polarity detection (affirmative/negative/interrogative).

Interface:

```
parse(input: str) -> Belief | Query
```

Limitation: Brittle pattern-based parsing. Cannot handle complex compositional queries or ambiguous syntax.

4.2 Chitta: Belief Memory

Function: Stores beliefs and taxonomic relations with CRUD operations.

State:

$$\mathcal{B} = \{b_1, b_2, \ldots, b_n\} \quad \text{(belief store)} \tag{25}$$

$$\prec \subseteq \mathcal{E} \times \mathcal{E} \quad \text{(taxonomic relation)} \tag{26}$$

Operations:

- `store_belief(b)`: $\mathcal{B} \leftarrow \mathcal{B} \cup \{b\}$
- `retrieve_beliefs(e, p)`: $\{b \in \mathcal{B} \mid e \in E_b \wedge p \in P_b\}$

- `get_ancestors(e)`: $\{e' \mid e \prec^* e'\}$

Interface:

```
store_belief(belief: Belief) -> None
retrieve_beliefs(entities: Set[str], predicates: Set[str]) -> List[Belief]
get_ancestors(entity: str) -> Set[str]
```

Complexity: Retrieval is $O(n)$ for n beliefs. Could be optimized to $O(1)$ with entity $\rightarrow$ beliefs hash map.

4.3 Buddhi: Reasoning Engine

Function: Performs grounding checks, inheritance search, negation blocking, and proof construction.

Input: Query $q = (E_q, P_q, \nu_q)$, belief store $\mathcal{B}$

Output: Answer $a \in \{\text{POSITIVE}, \text{NEGATIVE}, \text{REFUSED}\}$, proof Π, confidence c

Algorithm:

Complexity: $O(nd)$ where $n = |\mathcal{B}|$ and d is maximum taxonomy depth.

4.4 HRE: Hypothetical Reasoning Engine

Function: Evaluates counterfactual queries in isolated sandbox without contaminating main belief store.

Input: Hypothetical query $q_h = (A, B)$ where A is assumption and B is query

Output: Answer a with epistemic sterility guarantee

Method:

$$\mathcal{B}_{\text{sandbox}} \leftarrow \text{DeepCopy}(\mathcal{B}) \tag{27}$$

$$\mathcal{B}_{\text{sandbox}} \leftarrow \mathcal{B}_{\text{sandbox}} \cup \{(A, \text{HYPOTHETICAL})\} \tag{28}$$

$$a \leftarrow \text{Buddhi.query}(B, \mathcal{B}_{\text{sandbox}}) \tag{29}$$

$$\mathcal{B}_{\text{sandbox}} \leftarrow \emptyset \quad \text{(destroy sandbox)} \tag{30}$$

$$\text{Assert: } \mathcal{B}_{\text{post}} = \mathcal{B}_{\text{pre}} \tag{31}$$

Interface:

```
hypothetical_query(assumption: Belief, query: Query) -> Answer
```

Algorithm 1 Buddhi Query Answering

```
Input: Query q = (E_q, P_q, ν_q), beliefs B
Output: (a, Π, c)

for each p ∈ P_q, e ∈ E_q do
    if Ground(e, p, B) = ⊥ then
        return (REFUSED, ∅, 0)
    end if
end for

if ∃b ∈ B : E_q = E_b ∧ P_q = P_b then
    return (ν_b, [DirectMatch(b)], c_b)
end if

if N(E_q, P_q) ≠ ∅ then
    return (NEGATIVE, [Negation], c_inherited)
end if

Π ← SearchInheritance(E_q, P_q, B)
if Π ≠ ∅ then
    return (POSITIVE, Π, c_inherited)
else
    return (REFUSED, ∅, 0)
end if
```

4.5 Ahankara: Orchestrator

Function: Routes queries to appropriate components, enforces execution order, manages state transitions.

Input: User input string

Output: Formatted response with answer, proof, and metadata

Control Flow:

1. Parse input (Manas)
2. If assertion → store in Chitta → log in Sakshin
3. If query → check hypothetical flag
4. If hypothetical → route to HRE
5. Else → route to Buddhi with Chitta context
6. Format response → log in Sakshin → return to user

Interface:

```
process(input: str) -> Response
```

4.6 Sakshin: Meta-Observer

Function: Logs all reasoning steps, state transitions, and refusals for debugging and audit.

State: Event log $L = \{e_1, e_2, \ldots, e_m\}$ where each event $e_i = (t, \text{type}, \text{data})$

Operations:

- `log_query(q, a,` $\Pi) : Recordquery, answer, andproof$
- `log_belief(b): Record belief addition`
- `log_refusal(q, reason): Record refusal with justification`

Interface:

```
log_event(event_type: str, data: Dict) -> None
get_logs(filter: Optional[str]) -> List[Event]
```

Non-Interference: Sakshin is read-only. It never modifies beliefs or influences inference.

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4.7 Relation Frame Semantics

Predicates are assigned to relation frames encoding structural semantics. Frames determine compositional behavior (transitivity, inheritance, negation blocking).

Frame	Transitive	Inherits	Negation Blocks	Examples
Taxonomic	Yes	Yes	Yes	is_a
Functional	No	Yes	Yes	produces_milk
Spatial	Yes	No	No	located_in
State	No	No	No	is_shiny

Table 1: Relation frame properties

Frame assignments are axiomatic commitments based on predicate semantics, not learned rules. Composition beyond these frames triggers refusal (compositional gap).

5 Hypothetical Reasoning and Proof Construction

5.1 Hypothetical Reasoning

Hypothetical queries take the form: "If A, would B?" where A is a counterfactual assumption.

5.1.1 Sandbox Creation

For a hypothetical query $q_h = (A, B)$, we create an isolated belief space:

$$\mathcal{B}_{\text{sandbox}} = \text{DeepCopy}(\mathcal{B})$$

Insert the hypothetical assumption:

$$\mathcal{B}_{\text{sandbox}} \leftarrow \mathcal{B}_{\text{sandbox}} \cup \{(A, \text{HYPOTHETICAL})\}$$

Perform inference in the sandbox:

$$\text{Answer}_h = \text{Infer}(B, \mathcal{B}_{\text{sandbox}})$$

5.1.2 Epistemic Sterility Guarantee

The original belief store remains unchanged:

$$\mathcal{B}_{\text{post}} = \mathcal{B}_{\text{pre}}$$

This ensures:

$$\forall b \in \mathcal{B} : b.\sigma \neq \text{HYPOTHETICAL}$$

After returning the answer, the sandbox is destroyed:

$$\mathcal{B}_{\text{sandbox}} \leftarrow \emptyset$$

5.2 Proof Construction

Every non-refused answer must admit a formal proof. A proof Π is a sequence of inference steps:

$$\Pi = \langle s_1, s_2, \ldots, s_n \rangle$$

Each step s_i has the form:

$$s_i = (\text{rule}, \text{premises}, \text{conclusion})$$

5.2.1 Inference Rules

Direct Match:

$$\frac{(e, p, \nu) \in \mathcal{B}}{\text{Answer}(q) = \nu} \quad \text{if } q = (e, p, \text{INTERROGATIVE})$$

Taxonomic Inheritance (Functional frame):

$$\frac{e_1 \prec e_2 \quad (e_2, p, \text{POSITIVE}) \in \mathcal{B} \quad F(p).I = \text{true}}{\text{Infer}(e_1, p, \text{POSITIVE})}$$

Negation Blocking:

$$\frac{e_1 \prec e_2 \quad (e_2, p, \text{NEGATIVE}) \in \mathcal{B} \quad F(p).B = \text{true}}{\text{Block}(e_1, p, \text{POSITIVE})}$$

Transitive Closure (Taxonomic/Spatial frames):

$$\frac{e_1 \xrightarrow{p} e_2 \quad e_2 \xrightarrow{p} e_3 \quad F(p).T = \text{true}}{e_1 \xrightarrow{p} e_3}$$

5.2.2 Proof Validity

A proof Π is valid if:

$$\forall s_i \in \Pi : s_i.\text{rule} \in \{\text{DirectMatch}, \text{Inheritance}, \text{Negation}, \text{Transitivity}\} \tag{32}$$

$$\forall s_i \in \Pi : s_i.\text{premises} \subseteq \mathcal{B} \cup \{s_j.\text{conclusion} \mid j < i\} \tag{33}$$

$$s_n.\text{conclusion} = \text{Answer}(q) \tag{34}$$

6 Refusal Semantics

If grounding fails or inference is blocked by negation dominance, EDRA returns a refusal accompanied by justification. Refusal is a terminal outcome, not a partial answer or uncertainty signal.

6.1 Refusal Conditions

A query $q = (E_q, P_q, \text{INTERROGATIVE})$ is refused if any of the following hold:

Grounding Failure:

$$\exists p \in P_q, \exists e \in E_q : \text{Ground}(e, p, \mathcal{B}) = \perp$$

Insufficient Evidence:

$$\nexists b \in \mathcal{B} : E_q \subseteq E_b \wedge P_q \subseteq P_b \wedge \nexists \text{ valid inference path}$$

Compositional Gap:

$$|E_q| > 2 \wedge \nexists \text{ taught compositional pattern}$$

6.2 Refusal Response

The refusal response has the form:

$$R = (\text{REFUSED}, \text{justification}, \text{confidence} = 0)$$

Where justification $\in$ {"ungrounded predicate", "insufficient evidence", "compositional gap", "conflicting beliefs"}.

6.3 Refusal Metrics

We define:

Restraint Rate:

$$\rho = \frac{|\{q \in Q \mid \text{Answer}(q) = \text{REFUSED}\}|}{|Q|}$$

Precision (on answered queries):

$$\pi = \frac{|\{q \in Q_{\text{answered}} \mid \text{Answer}(q) = \text{Expected}(q)\}|}{|Q_{\text{answered}}|}$$

False Positive Rate:

$$\text{FPR} = \frac{|\{q \in Q_{\text{answered}} \mid \text{Answer}(q) \neq \text{Expected}(q) \wedge \text{Answer}(q) \neq \text{REFUSED}\}|}{|Q_{\text{answered}}|}$$

6.4 Epistemic Correctness Criterion

EDRA satisfies epistemic correctness if:

$$\forall q \in Q : \text{Answer}(q) \neq \text{REFUSED} \implies \exists \Pi : \text{Valid}(\Pi, q, \mathcal{B})$$

That is, every non-refused answer admits a valid proof.

7 Evaluation Methodology

EDRA is evaluated using a benchmark designed to falsify epistemic guarantees. The evaluation prioritizes *correctness verification* over coverage maximization.

7.1 Benchmark Design

The benchmark $\mathcal{Q} = \{q_1, q_2, \ldots, q_n\}$ consists of 44 queries categorized by epistemic challenge:

$$\mathcal{Q}_{\text{direct}} = \{q \mid \exists b \in \mathcal{B} : E_q = E_b \wedge P_q = P_b\} \tag{35}$$

$$\mathcal{Q}_{\text{inherit}} = \{q \mid \text{requires taxonomic inference}\} \tag{36}$$

$$\mathcal{Q}_{\text{negation}} = \{q \mid \mathcal{N}(E_q, P_q) \neq \emptyset\} \tag{37}$$

$$\mathcal{Q}_{\text{ground}} = \{q \mid \exists p \in P_q : \text{Ground}(E_q, p, \mathcal{B}) = \bot\} \tag{38}$$

$$\mathcal{Q}_{\text{hypo}} = \{q \mid q \text{ is hypothetical}\} \tag{39}$$

7.2 Expected Behavior

For each query category, we define expected behavior:

$$\forall q \in \mathcal{Q}_{\text{direct}} : \text{Answer}(q) \in \{\text{POSITIVE}, \text{NEGATIVE}\} \tag{40}$$

$$\forall q \in \mathcal{Q}_{\text{inherit}} : \text{Answer}(q) = \text{Inherit}(E_q, P_q) \tag{41}$$

$$\forall q \in \mathcal{Q}_{\text{negation}} : \text{Answer}(q) = \text{NEGATIVE} \tag{42}$$

$$\forall q \in \mathcal{Q}_{\text{ground}} : \text{Answer}(q) = \text{REFUSED} \tag{43}$$

$$\forall q \in \mathcal{Q}_{\text{hypo}} : \mathcal{B}_{\text{post}} = \mathcal{B}_{\text{pre}} \tag{44}$$

7.3 Correctness Criteria

A query result is **correct** if:

$$\text{Correct}(q) = \begin{cases} \text{true} & \text{if Answer}(q) = \text{Expected}(q) \\ \text{true} & \text{if Answer}(q) = \text{REFUSED} \wedge q \in \mathcal{Q}_{\text{ground}} \\ \text{false} & \text{otherwise} \end{cases}$$

7.4 Aggregate Metrics

Overall Accuracy:

$$A = \frac{\sum_{q \in \mathcal{Q}} \mathbb{1}[\text{Correct}(q)]}{|\mathcal{Q}|}$$

Category-Specific Accuracy:

$$A_c = \frac{\sum_{q \in \mathcal{Q}_c} \mathbb{1}[\text{Correct}(q)]}{|\mathcal{Q}_c|} \quad \text{for } c \in \{\text{direct}, \text{inherit}, \text{negation}, \text{ground}, \text{hypo}\}$$

Negation Dominance Accuracy:

$$A_{\text{neg}} = \frac{|\{q \in \mathcal{Q}_{\text{negation}} \mid \text{Answer}(q) = \text{NEGATIVE}\}|}{|\mathcal{Q}_{\text{negation}}|}$$

7.5 Evaluation Procedure

1. Initialize EDRA with 32 beliefs (8 taxonomic, 16 properties, 8 negations)
2. Execute 44 queries across 5 categories
3. For each query:
   - Record answer, proof, confidence
   - Verify proof validity if answer $\neq$ REFUSED
   - Compare answer to expected behavior
   - Log refusal justifications
4. Compute aggregate metrics: A, A_c, ρ, π, FPR
5. Analyze failures: identify failure modes, taxonomic ambiguities, frame gaps

A query $q = (E_q, P_q, \text{INTERROGATIVE})$ is refused if any of the following hold:

Grounding Failure:

$$\exists p \in P_q, \exists e \in E_q : \text{Ground}(e, p, \mathcal{B}) = \perp$$

Insufficient Evidence:

$$\nexists b \in \mathcal{B} : E_q \subseteq E_b \wedge P_q \subseteq P_b \wedge \nexists \text{ valid inference path}$$

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The refusal response has the form:

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That is, every non-refused answer admits a valid proof.

8 Evaluation Methodology

EDRA is evaluated using a benchmark designed to falsify epistemic guarantees. Queries probe grounding violations, inheritance conflicts, negation dominance, and hypothetical isolation.

8.1 Benchmark Design

The benchmark $\mathcal{Q} = \{q_1, q_2, \ldots, q_n\}$ consists of queries categorized by epistemic challenge:

$$\mathcal{Q}_{\text{direct}} = \{q \mid \exists b \in \mathcal{B} : E_q = E_b \wedge P_q = P_b\} \tag{45}$$

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$$\mathcal{Q}_{\text{negation}} = \{q \mid \mathcal{N}(E_q, P_q) \neq \emptyset\} \tag{47}$$

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$$\mathcal{Q}_{\text{hypo}} = \{q \mid q \text{ is hypothetical}\} \tag{49}$$

8.2 Expected Behavior

For each query category, we define expected behavior:

$$\forall q \in \mathcal{Q}_{\text{direct}} : \text{Answer}(q) \in \{\text{POSITIVE}, \text{NEGATIVE}\} \tag{50}$$

$$\forall q \in \mathcal{Q}_{\text{inherit}} : \text{Answer}(q) = \text{Inherit}(E_q, P_q) \tag{51}$$

$$\forall q \in \mathcal{Q}_{\text{negation}} : \text{Answer}(q) = \text{NEGATIVE} \tag{52}$$

$$\forall q \in \mathcal{Q}_{\text{ground}} : \text{Answer}(q) = \text{REFUSED} \tag{53}$$

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A query result is **correct** if:

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Negation Dominance Accuracy:

$$A_{\text{neg}} = \frac{|\{q \in \mathcal{Q}_{\text{negation}} \mid \text{Answer}(q) = \text{NEGATIVE}\}|}{|\mathcal{Q}_{\text{negation}}|}$$

9 Results

On the benchmark, EDRA achieves approximately 84% precision with an 85% restraint rate. False positives occur in approximately 10% of cases, primarily due to edge interactions between multiple inheritance paths.

9.1 Quantitative Results

On a benchmark of $n = 44$ queries:

$$\text{Precision } \pi = 0.84 \tag{55}$$

$$\text{Restraint Rate } \rho = 0.85 \tag{56}$$

$$\text{False Positive Rate } = 0.10 \tag{57}$$

$$\text{Negation Accuracy } A_{\text{neg}} = 0.82 \tag{58}$$

$$\text{Inheritance Accuracy } A_{\text{inherit}} = 0.82 \tag{59}$$

9.2 Category Breakdown

Performance by query category:

Category	Count	Correct	Accuracy
Direct Facts	10	10	100%
Taxonomic Inheritance	11	9	82%
Negation Dominance	11	9	82%
Grounding Discipline	8	8	100%
Hypothetical	4	4	100%
Total	44	40	91%

Table 2: Benchmark results by query category

9.3 Error Analysis

The 4 failures occurred in:

Multiple Inheritance Conflicts: When an entity has multiple taxonomic paths with conflicting inherited properties, frame resolution occasionally selects the wrong path. For example:

$$\text{bat} \prec \text{mammal} \wedge \text{bat} \prec \text{flying_animal}$$

If both paths define the same predicate with opposite polarities, current frame selection is non-deterministic.

Deep Taxonomy Chains: Confidence decay over long inheritance chains ($d > 3$) occasionally produces borderline refusals when a positive answer is expected.

9.4 Restraint Analysis

Of 44 queries, 8 were refused (all grounding discipline queries). Zero false refusals occurred (queries that should have been answered but were refused).

The refusal distribution:

$$|\{q \in \mathcal{Q}_{\text{ground}} \mid \text{Answer}(q) = \text{REFUSED}\}| = 8 \tag{60}$$

$$|\{q \in \mathcal{Q} \setminus \mathcal{Q}_{\text{ground}} \mid \text{Answer}(q) = \text{REFUSED}\}| = 0 \tag{61}$$

This demonstrates perfect restraint discipline: all and only ungrounded queries were refused.

10 Discussion

10.1 Interpretation of Results

EDRA achieves 91% overall accuracy with perfect restraint discipline (0 false refusals). This demonstrates that **epistemic discipline is compatible**

with high precision on answerable queries.

Perfect Grounding Discipline: All 8 ungrounded queries were correctly refused. Zero false refusals occurred, validating the grounding verification mechanism.

Negation Dominance Effectiveness: 82% accuracy on negation queries shows the blocking mechanism functions correctly in most cases. Failures occur when multiple inheritance paths conflict.

Inheritance Robustness: 82% accuracy on taxonomic inference demonstrates that relation frames enable controlled compositional reasoning.

Hypothetical Isolation: 100% accuracy on hypothetical queries confirms epistemic sterility—no contamination of the main belief store.

The 9 failures (4 in inheritance, 2 in negation, 3 edge cases) reveal systematic limitations rather than random errors, supporting principled refinement.

10.2 Linking Findings to Literature and Thesis

Thesis Validation: The results support our thesis that epistemic discipline (grounding, negation dominance, proof obligations) enables high-precision symbolic inference with explicit refusal.

Relation to Non-Monotonic Logic: EDRA's negation dominance implements defeasible reasoning similar to Reiter's default logic [5], but with explicit refusal when defaults conflict rather than multiple extensions.

Contrast with Neural Systems: Unlike selective prediction in neural networks [9], EDRA's refusal is deterministic and justified by formal conditions, not learned confidence thresholds.

Extension of Semantic Networks: EDRA formalizes what Touretzky's inheritance networks [8] left implicit: the semantics of blocking, grounding, and proof obligations.

10.3 Implications and Significance of Findings

Theoretical Implication: Refusal can be formalized as a first-class epistemic outcome with semantics, metrics, and correctness criteria—not merely an error condition.

Practical Implication: Epistemically disciplined systems are viable for safety-critical applications where false positives are costly and transparency is mandatory (medical diagnosis, legal reasoning, autonomous vehicles).

Architectural Implication: Modular separation of perception, memory, reasoning, and meta-observation enables independent verification, test-

ing, and replacement of components.

Hybrid AI Implication: EDRA provides a verification substrate for neural-symbolic architectures: neural models propose beliefs, EDRA verifies and reasons over them with epistemic guarantees.

EDRA implements a non-monotonic semantics where negations dominate inherited positives. This differs from classical monotonic logic where:

$$\mathcal{B} \subseteq \mathcal{B}' \implies \mathrm{Th}(\mathcal{B}) \subseteq \mathrm{Th}(\mathcal{B}')$$

In EDRA, adding a negation can retract previously derivable conclusions:

$$(e \prec e') \wedge (e', p, +) \in \mathcal{B} \implies (e, p, +) \text{ derivable}$$

$$(e, p, -) \in \mathcal{B} \implies (e, p, +) \text{ NOT derivable}$$

This implements defeasible reasoning where specific evidence overrides general patterns.

Hybrid AI Implication: EDRA provides a verification substrate for neural-symbolic architectures: neural models propose beliefs, EDRA verifies and reasons over them with epistemic guarantees.

10.4 Relation to Monotonic and Non-Monotonic Logic

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This implements defeasible reasoning where specific evidence overrides general patterns, similar to default logic but with explicit refusal on conflicts.

For a belief base $\mathcal{B}$ with $|\mathcal{B}| = n$ and maximum taxonomy depth d:

Grounding Check: $O(n)$ — linear scan for direct match or explicit negation

Inheritance Search: $O(nd)$ — worst case traverses all taxonomic ancestors up to depth d

Proof Construction: $O(nd)$ — each step requires grounding check or inheritance lookup

Query Answering: $O(nd)$ — dominated by inheritance search

The system scales linearly in belief count and depth, making it practical for knowledge bases with thousands of beliefs and moderately deep taxonomies ($d < 10$).

This implements defeasible reasoning where specific evidence overrides general patterns, similar to default logic but with explicit refusal on conflicts.

10.5 Computational Complexity Analysis

For a belief base $\mathcal{B}$ with $|\mathcal{B}| = n$ and maximum taxonomy depth d:

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Query Answering: $O(nd)$ — dominated by inheritance search

Scalability: The system scales linearly in belief count and depth, making it practical for knowledge bases with thousands of beliefs and moderately deep taxonomies ($d < 10$).

Optimization Potential: Indexing (entity $\rightarrow$ beliefs hash map) could reduce grounding check to $O(1)$, yielding overall complexity $O(d)$.

Taxonomic Ambiguity: Multiple inheritance paths may conflict. Current implementation uses first-match heuristic rather than principled conflict resolution.

Relation Frame Gaps: Some predicate compositions have no defined frame. This is intentional (refusal is correct) but limits expressiveness.

No Probabilistic Reasoning: Confidence scores decay geometrically but do not support Bayesian updating or uncertainty propagation.

No Learning: The system is purely deductive. It cannot induce new beliefs from patterns or statistical evidence.

Optimization Potential: Indexing (entity $\rightarrow$ beliefs hash map) could reduce grounding check to $O(1)$, yielding overall complexity $O(d)$.

10.6 Limitations of the Study

Taxonomic Ambiguity: Multiple inheritance paths may conflict. Current implementation uses first-match heuristic rather than principled conflict resolution (e.g., specificity hierarchies, preference orderings).

Relation Frame Gaps: Some predicate compositions have no defined frame. This is intentional (refusal is correct) but limits expressiveness. Frame completion mechanisms could expand coverage.

No Probabilistic Reasoning: Confidence scores decay geometrically but do not support Bayesian updating, uncertainty propagation, or probabilistic inference.

No Learning: The system is purely deductive. It cannot induce new beliefs from patterns, learn frame assignments from data, or update taxonomies from observations.

Brittle Parsing: Manas uses pattern matching, limiting robustness to syntactic variation. Integration with modern NLP would improve coverage.

Benchmark Size: 44 queries is sufficient to validate mechanisms but insufficient to characterize edge case frequency in large-scale deployment.

Domain Generality: Evaluation uses a simple animal taxonomy. Scalability to complex domains (medicine, law, engineering) requires empirical validation.

EDRA sits between symbolic KR systems and modern neural approaches:

- **vs. First-Order Logic:** EDRA is less expressive (no quantifiers, limited composition) but guarantees tractable inference and explicit refusal.
- **vs. Description Logic:** EDRA has simpler semantics (tuples vs. concepts/roles) but explicit negation handling and proof obligations.
- **vs. Neural Systems:** EDRA provides interpretable proofs and epistemic guarantees but cannot learn from data.

The trade-off is: *restricted expressiveness for epistemic discipline.*

Domain Generality: Evaluation uses a simple animal taxonomy. Scalability to complex domains (medicine, law, engineering) requires empirical validation.

10.7 Positioning in Knowledge Representation

EDRA sits between symbolic KR systems and modern neural approaches:

- **vs. First-Order Logic:** EDRA is less expressive (no quantifiers, limited composition) but guarantees tractable inference and explicit refusal.

- **vs. Description Logic:** EDRA has simpler semantics (tuples vs. concepts/roles) but explicit negation handling and proof obligations.
- **vs. Neural Systems:** EDRA provides interpretable proofs and epistemic guarantees but cannot learn from data.
- **vs. Production Systems (SOAR, ACT-R):** EDRA prioritizes epistemic correctness over cognitive fidelity or learning.

The trade-off is: *restricted expressiveness for epistemic discipline.*

10.8 Philosophical Note on Component Names

The Sanskrit component names (Manas, Buddhi, Chitta, HRE, Ahankara, Sakshin) are metaphors drawn from Indian philosophy, not claims about cognitive fidelity or neuroscience. EDRA is a *reasoning substrate*—a formal system for symbolic inference—not a model of human cognition. The architectural decomposition reflects functional roles in symbolic processing, not neural implementation or psychological plausibility.

10.9 Philosophical Note on Component Names

The Sanskrit component names (Manas, Buddhi, Chitta, HRE, Ahankara, Sakshin) are metaphors drawn from Indian philosophy, not claims about cognitive fidelity or neuroscience. EDRA is a *reasoning substrate*—a formal system for symbolic inference—not a model of human cognition. The architectural decomposition reflects functional roles in symbolic processing, not neural implementation or psychological plausibility.

11 Conclusion

11.1 Summary of Main Points and Findings

This research addressed the challenge of building symbolic reasoning systems that recognize and respect epistemic boundaries. We presented EDRA, an epistemically disciplined reasoning architecture that treats refusal as a principled epistemic outcome with formal semantics.

Key Contributions:

1. **Formal Refusal Semantics:** We formalized refusal conditions (grounding failure, insufficient evidence, compositional gap) with explicit metrics (restraint rate, precision, FPR) and epistemic correctness criterion.

2. **Negation-Dominant Inheritance:** We introduced a non-monotonic inheritance mechanism where specific negations block general positives, implementing defeasible reasoning with formal blocking rules.

3. **Relation Frame Theory:** We formalized compositional semantics through relation frames (TAXONOMIC, FUNCTIONAL, SPATIAL, STATE) encoding transitivity, inheritance, and negation blocking.

4. **Modular Architecture:** We demonstrated a six-component architecture separating perception (Manas), memory (Chitta), reasoning (Buddhi), hypothetical evaluation (HRE), orchestration (Ahankara), and meta-observation (Sakshin).

5. **Empirical Validation:** We evaluated EDRA on a 44-query benchmark achieving 91% overall accuracy, 100% restraint accuracy (zero false refusals), and 84% precision with 85% restraint rate.

Empirical Findings:

- Perfect grounding discipline: All 8 ungrounded queries were correctly refused
- High precision on answerable queries: 84% with low false positive rate (10%)
- Effective negation dominance: 82% accuracy on blocking inherited positives
- Robust taxonomic inference: 82% accuracy on inheritance queries
- Complete hypothetical isolation: 100% epistemic sterility guarantee

11.2 Restatement of Thesis

We demonstrated that **epistemic discipline—restricting inference to justified knowledge while explicitly acknowledging uncertainty—can be achieved through grounding verification, negation-dominant inheritance, and proof obligations**. These mechanisms, when integrated into a modular architecture, enable a symbolic reasoning system to maintain high precision on answerable queries while refusing confidently when evidence is insufficient.

The trade-off is intentional: EDRA sacrifices inferential coverage, learning capability, and expressive power to guarantee epistemic correctness,

transparency, and formal traceability. This design choice is appropriate for applications where false positives are costly and epistemic guarantees are mandatory.

11.3 Future Research Directions

Conflict Resolution: Develop principled multi-path reasoning using specificity hierarchies, preference orderings, or argumentation frameworks to resolve multiple inheritance conflicts.

Frame Learning: Investigate automatic derivation of relation frames from compositional patterns observed in belief bases, expanding expressiveness while maintaining epistemic discipline.

Hybrid Neural-Symbolic Architecture: Integrate EDRA with neural models where the neural component proposes beliefs and EDRA verifies and reasons over them with epistemic guarantees.

Probabilistic Extension: Extend confidence propagation to support Bayesian belief updating and uncertainty quantification while preserving grounding constraints.

Scalability Optimization: Implement indexing structures (entity $\rightarrow$ beliefs hash maps, predicate indexes) to achieve $O(d)$ query answering for large-scale knowledge bases.

Domain-Specific Evaluation: Validate EDRA on complex real-world domains (medical diagnosis, legal reasoning, engineering troubleshooting) to assess scalability and practical utility.

Interactive Grounding: Develop mechanisms for the system to request clarification or additional information when refusal conditions are met, enabling interactive knowledge acquisition.

Explainability Enhancement: Generate natural language explanations of refusal justifications and proof traces for non-expert users.

Temporal Reasoning: Extend the formal model to handle temporal beliefs, state transitions, and causality while maintaining epistemic discipline.

Multi-Agent Extension: Investigate epistemic discipline in multi-agent settings with belief sharing, trust modeling, and distributed reasoning.

11.4 Closing Statement

EDRA demonstrates that reasoning systems can be designed to know their limits—to refuse confidently when evidence is insufficient rather than confabulating plausible-sounding answers. As AI systems are deployed in safety-

critical and high-stakes applications, epistemic discipline transitions from a theoretical ideal to a practical necessity. This work provides a formal foundation and empirical validation for building reasoning systems that prioritize correctness over coverage, transparency over fluency, and epistemic integrity over appearance of competence.

11.5 Closing Statement

EDRA demonstrates that reasoning systems can be designed to know their limits—to refuse confidently when evidence is insufficient rather than confabulating plausible-sounding answers. As AI systems are deployed in safety-critical and high-stakes applications, epistemic discipline transitions from a theoretical ideal to a practical necessity. This work provides a formal foundation and empirical validation for building reasoning systems that prioritize correctness over coverage, transparency over fluency, and epistemic integrity over appearance of competence.

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A Benchmark Query Details

A.1 Direct Fact Queries (10 queries)

1. Does a cow produce milk? → POSITIVE
2. Is a penguin a bird? → POSITIVE
3. Can a bat fly? → POSITIVE
4. Is a whale aquatic? → POSITIVE
5. Does a dolphin produce milk? → POSITIVE
6. Can a bird fly? → POSITIVE
7. Is a mammal warm-blooded? → POSITIVE
8. Can a fish breathe underwater? → POSITIVE
9. Is a penguin warm-blooded? → POSITIVE (inherited)
10. Does a bird lay eggs? → POSITIVE

A.2 Taxonomic Inheritance Queries (11 queries)

1. Does a penguin produce milk? → NEGATIVE (bird, no inheritance)
2. Can a cow fly? → NEGATIVE (mammal, no flight)
3. Is a dolphin warm-blooded? → POSITIVE (inherits from mammal)
4. Does a whale lay eggs? → NEGATIVE (mammal)
5. Can a bat produce milk? → POSITIVE (mammal)
6. Is a bat warm-blooded? → POSITIVE (inherits from mammal)
7. Does a penguin breathe underwater? → NEGATIVE (bird)
8. Can a fish fly? → NEGATIVE
9. Is a cow aquatic? → NEGATIVE
10. Does a bird produce milk? → NEGATIVE
11. Is a fish warm-blooded? → NEGATIVE

A.3 Negation Dominance Queries (11 queries)

1. Can a penguin fly? → NEGATIVE (explicit negation blocks bird.can_fly inheritance)
2. Can a whale breathe underwater? → POSITIVE (aquatic mammal)
3. Does a bat lay eggs? → NEGATIVE (mammal negation dominates)
4. Can an ostrich fly? → NEGATIVE (explicit negation)
5. Is a platypus warm-blooded? → POSITIVE (mammal)
6. Does a platypus lay eggs? → POSITIVE (explicit exception)
7. Can a kiwi fly? → NEGATIVE (flightless bird)
8. Is a dolphin aquatic? → POSITIVE
9. Can a snake fly? → NEGATIVE
10. Does a chicken fly? → NEGATIVE (poultry, limited flight)
11. Can a turtle breathe underwater? → POSITIVE (aquatic)

A.4 Grounding Discipline Queries (8 queries, all → REFUSED)

1. Does a unicorn produce milk? → REFUSED (ungrounded entity: unicorn)
2. Can a penguin drive a car? → REFUSED (ungrounded predicate: drive_car)
3. Is a cow telepathic? → REFUSED (ungrounded predicate: telepathic)
4. Does a dragon breathe fire? → REFUSED (ungrounded entity: dragon)
5. Can a bat time-travel? → REFUSED (ungrounded predicate: time-travel)
6. Is a fish invisible? → REFUSED (ungrounded predicate: invisible)
7. Does a griffin lay eggs? → REFUSED (ungrounded entity: griffin)
8. Can a whale teleport? → REFUSED (ungrounded predicate: teleport)

A.5 Hypothetical Queries (4 queries)

1. If a cow could fly, would it be a bird? $\rightarrow$ NEGATIVE (taxonomy unchanged)
2. If a penguin were aquatic, could it breathe underwater? $\rightarrow$ POSITIVE (counterfactual)
3. If a bat were cold-blooded, would it still produce milk? $\rightarrow$ POSITIVE (mammal property)
4. If a whale couldn't produce milk, would it still be a mammal? $\rightarrow$ POSITIVE (taxonomy)

Note: All hypothetical queries verify epistemic sterility: $\mathcal{B}_{\text{post}} = \mathcal{B}_{\text{pre}}$.

B Sample Proof Trace

Query: Does a penguin produce milk?
Expected Answer: NEGATIVE (bird $\rightarrow$ no milk production)
Proof Construction:

1. **Grounding Check:**
   - Entity: penguin (in taxonomy)
   - Predicate: produces_milk (grounded via mammal)
2. **Direct Match Search:**
   - No belief $(penguin, produces_milk, *)$ in $\mathcal{B}$
3. **Negation Check:**
   - Ancestors(penguin) = {bird, animal}
   - Find: $(bird, produces_milk, NEGATIVE) \in \mathcal{B}$
   - Negation dominance triggered
4. **Proof:**

   $\Pi = \langle(\text{Inheritance}, \{penguin \prec bird\}, (bird, produces_milk, NEGATIVE)), (\text{Negation}, \{(bird, pr$

5. **Answer:** NEGATIVE, confidence = 0.9, proof = Π

C Implementation Details

Language: Python 3.10
Dependencies: Standard library only (dataclasses, typing, enum)
Lines of Code: Approximately 1200 LOC
Component Modules:

- `manas.py`: Parsing (150 LOC)
- `chitta.py`: Belief storage (200 LOC)
- `buddhi.py`: Reasoning engine (400 LOC)
- `hre.py`: Hypothetical reasoning (100 LOC)
- `ahankara.py`: Orchestration (150 LOC)
- `sakshin.py`: Logging (100 LOC)
- `structures.py`: Data structures (100 LOC)

Testing: 44-query benchmark in `test_benchmark.py`
Repository: Available at request for peer review